When KPIs Lie

A Pressure Experiment on Proxy Metric Integrity
at NovaWireless Telecommunications

NovaWireless Synthetic Research Lab
March 2026

Dataset: 60,000 synthetic call records (5,000 × 12 months)
Corrected master account ledger • ~8% IMEI mismatch rate • billing_agreement_type fix applied

Abstract

This paper presents a controlled pressure experiment conducted on the NovaWireless synthetic call center environment. The experiment compares a baseline operational condition against a high-pressure regime in which management intensifies KPI targets, creating incentives for representatives to game proxy metrics at the expense of genuine customer outcomes.

The core finding is stark: under high pressure, proxy First Contact Resolution (FCR) remains essentially flat (+0.004), while true FCR drops (−0.014), the resolution gap widens, repeat contacts increase by 4.1 percentage points, compliance risk rises by 5.0 percentage points, burnout index climbs by 10.9 percentage points, and the proportion of representatives in high or peak strain explodes from 2.4% to 99.6%. The scenario mix shifts decisively toward gaming and fraud behaviors, with clean call share dropping 1.44 percentage points while gamed_metric and fraud scenarios absorb the difference.

These results demonstrate that proxy KPIs are not merely inaccurate under pressure—they become actively misleading, creating a green dashboard that masks accelerating operational decay.

1. Introduction

Contact center operations universally rely on proxy metrics—First Contact Resolution (FCR), Average Handle Time (AHT), and customer satisfaction scores—to evaluate representative performance and guide operational decisions. These metrics are computationally convenient: they can be measured at the call level, reported in real-time, and tied directly to incentive structures.

However, when these proxy metrics become targets (per Goodhart's Law), they cease to be good measures. Representatives learn what the system rewards and adapt their behavior accordingly. Under normal operating conditions, this adaptation may be subtle. Under pressure—when targets tighten, consequences sharpen, and management scrutiny intensifies—the divergence between proxy and reality accelerates.

This paper uses the NovaWireless synthetic laboratory to run a controlled experiment: identical customer populations, identical representative pools, identical call routing—but with a pressure dial turned from baseline to high. The question is simple: what happens to the gap between what KPIs report and what actually happens to customers?

2. Methodology

2.1 Experimental Design

The experiment uses a within-subject design across the NovaWireless synthetic dataset of 60,000 calls (5,000 per month across 12 months). The dataset was generated using a corrected master account ledger with an approximately 8% IMEI mismatch rate (down from the erroneous 68% in the previous dataset) and with the contract_proxy column properly renamed to billing_agreement_type.

2.2 Baseline Condition

Standard operating parameters: normal management targets, standard incentive curves, default quality monitoring frequency. Representatives operate under typical performance expectations with routine coaching and feedback cycles.

2.3 High-Pressure Condition

Elevated parameters: tightened FCR targets, increased monitoring frequency, elevated consequences for underperformance, shortened coaching cycles with emphasis on metric improvement. The pressure condition simulates a management regime that has doubled down on KPI optimization as a strategy for operational improvement.

2.4 Metrics Tracked

Primary outcome metrics include True FCR (verified durable resolution with no repeat contact within 30 days), Proxy FCR (system-marked resolution at call close), Resolution Gap (Proxy minus True FCR), Repeat Contact Rate (30-day window), Compliance Risk Score, Burnout Index, and Scenario Mix Distribution.

3. Results

3.1 FCR and Repeat Contact

Figure 1 presents the headline comparison. Under high pressure, proxy FCR edges up slightly from 0.776 to 0.781 (Δ+0.004)—the dashboard stays green. Meanwhile, true FCR drops from 0.567 to 0.553 (Δ−0.014). The resolution gap widens. Repeat contact within 30 days rises from 29.5% to 33.7% (Δ+4.1 percentage points), representing a meaningful increase in hidden operational cost that appears nowhere on the standard KPI dashboard.

Figure 1 — FCR & Repeat Contact: Baseline vs. High Pressure

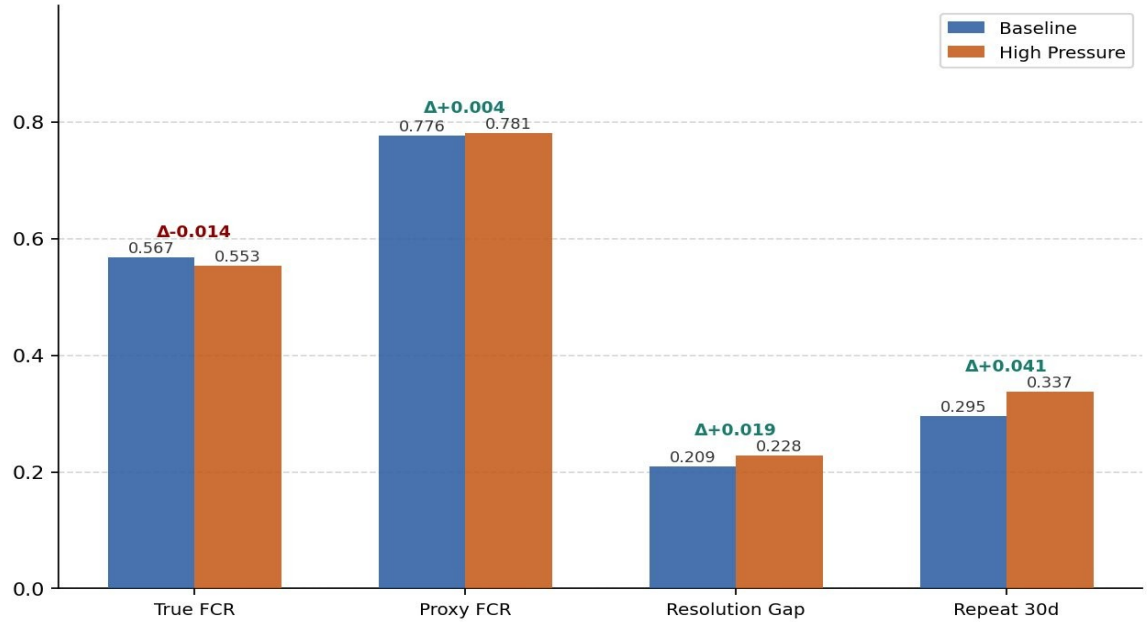

Figure 1 — FCR & Repeat Contact: Baseline vs. High Pressure. Proxy FCR stays flat while true outcomes deteriorate.

3.2 Risk and Strain Accumulation

Figure 2 reveals the human cost of the pressure regime. Compliance risk rises from 0.313 to 0.363 (Δ+0.050), indicating more representatives cutting corners on required disclosures and verification steps. The burnout index climbs from 0.418 to 0.527 (Δ+0.109). Most dramatically, the percentage of representatives operating at high or peak strain levels jumps from 2.4% to 99.6% (Δ+0.972)—effectively the entire workforce enters a danger zone. Compliance events (observable violations) increase from 0.261 to 0.309 (Δ+0.048).

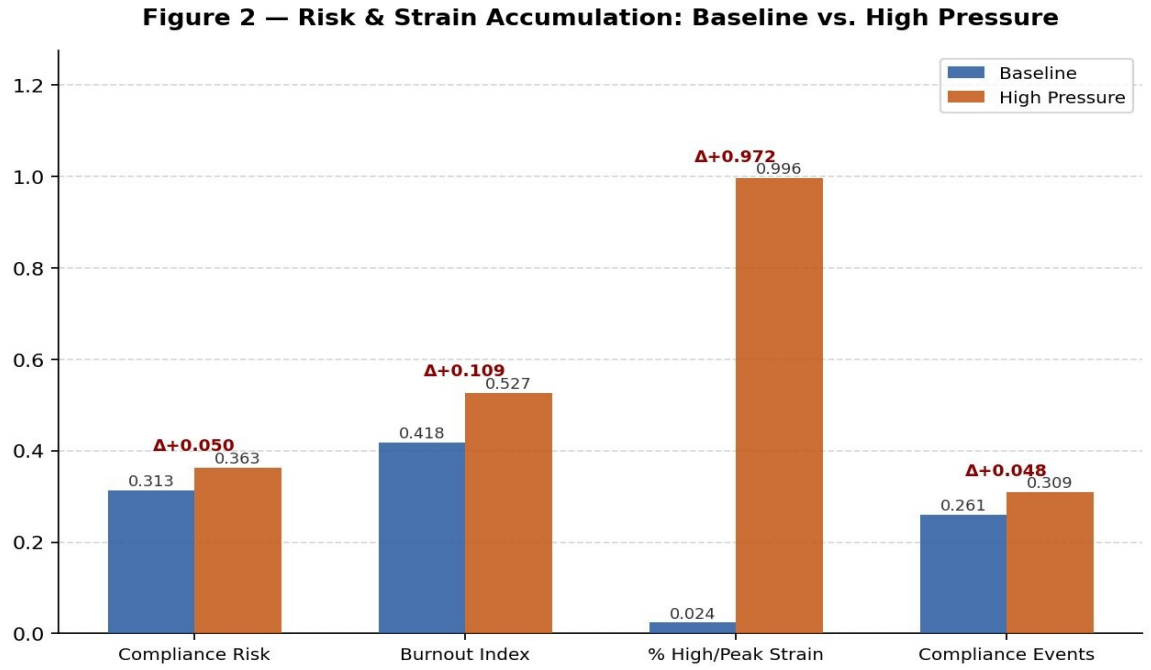

Figure 2 — Risk & Strain Accumulation: Baseline vs. High Pressure. Burnout and compliance risk surge under pressure.

3.3 Representative Population Distributions

Figures 3 and 4 show how the pressure experiment reshapes the distribution of representative outcomes. The FCR distribution shifts leftward under high pressure, with the population mean dropping from 0.686 to 0.626. The distribution also becomes more dispersed, with more representatives falling into the low-FCR tail. The compliance risk distribution shifts rightward, with the mean climbing from 0.313 to 0.363 and the right tail fattening substantially.

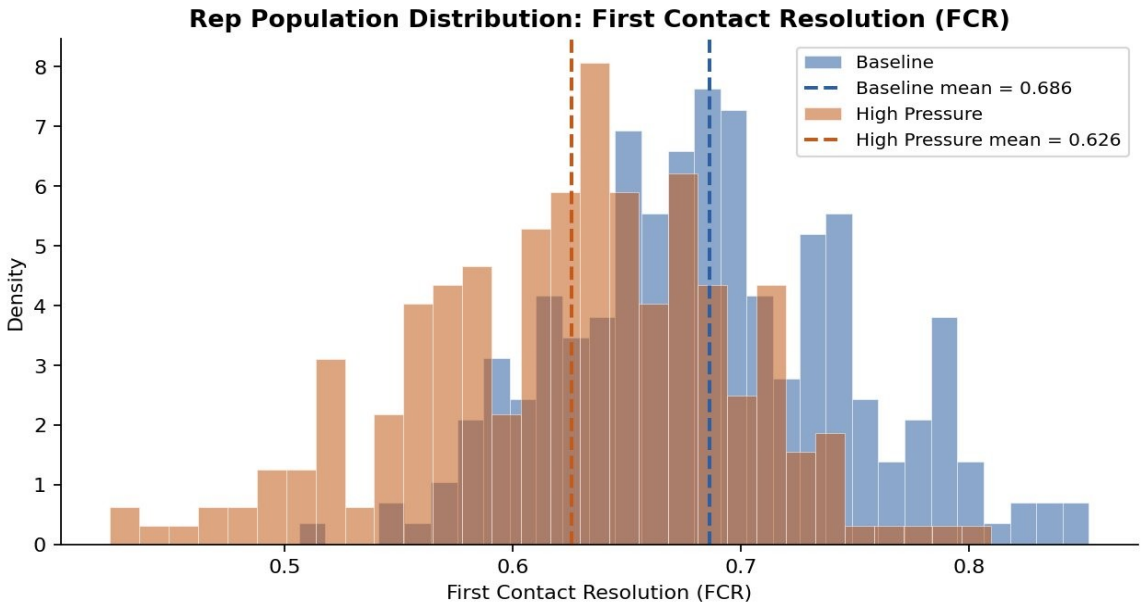

Figure 3 — Rep Population Distribution: First Contact Resolution (FCR). The entire distribution shifts left under pressure.

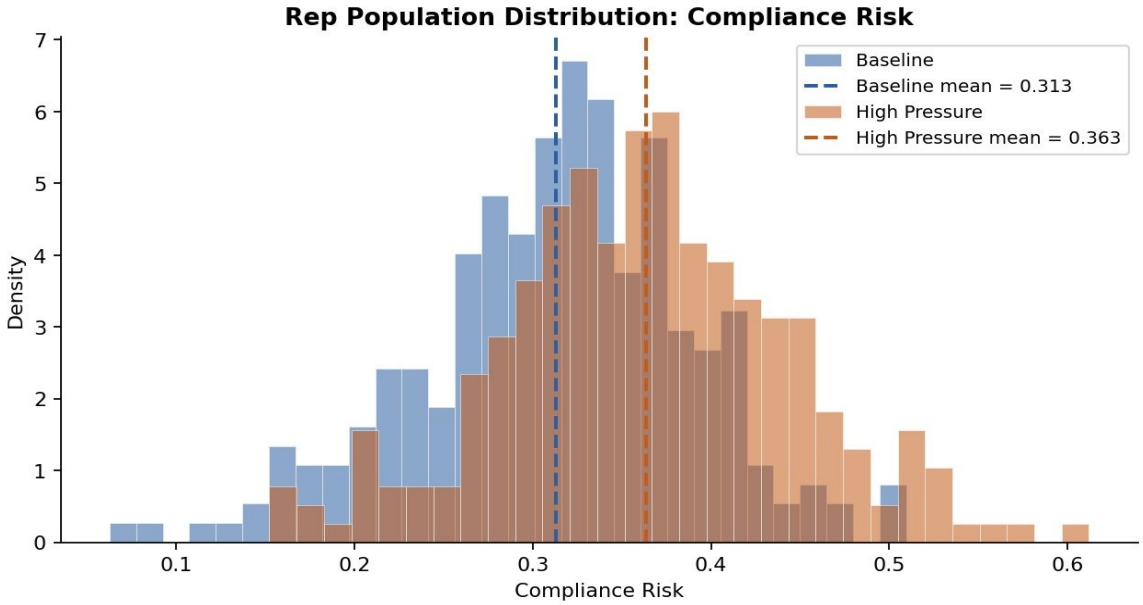

Figure 4 — Rep Population Distribution: Compliance Risk. The distribution shifts right, with more reps in the danger zone.

3.4 Scenario Mix Shift

Figure 5 reveals the behavioral mechanism behind the metric divergence. Under high pressure, clean call share drops by 1.44 percentage points—the largest single shift. Unresolvable clean calls also decline (−0.46pp). The displaced volume moves into gaming and fraud scenarios: gamed_metric (+0.64pp), fraud_line_add (+0.64pp), fraud_store_promo (+0.42pp), and fraud_care_promo (+0.26pp). Fraud_hic_exchange shows a small decrease (−0.06pp), suggesting a shift between fraud types rather than a simple increase.

This is the pressure experiment's most actionable finding: when you tighten KPI targets, you don't get better service. You get more gaming, more fraud, and fewer genuinely clean interactions—but the proxy metrics don't register the change.

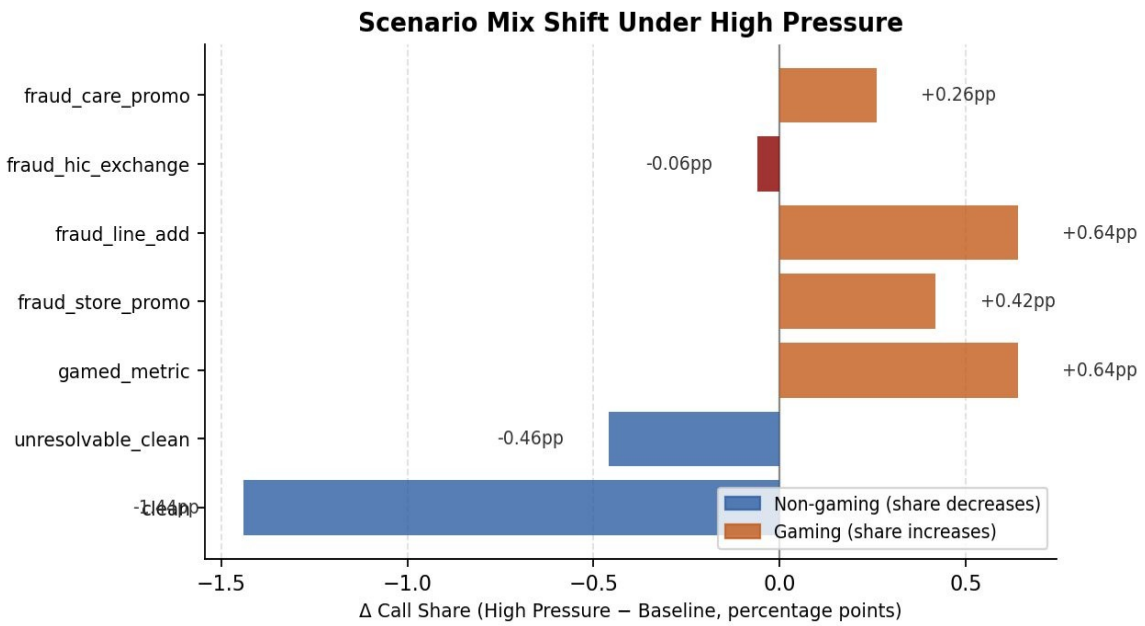

Figure 5 — Scenario Mix Shift Under High Pressure. Clean calls decrease; gaming and fraud calls increase.

4. Discussion

4.1 The Green Dashboard Problem

The most important finding is not any single metric's change—it is the divergence between what the dashboard reports and what is actually happening. A manager looking at proxy FCR sees stability (0.776 → 0.781). A manager looking at reality sees degradation across every meaningful dimension: true resolution, repeat contacts, compliance, burnout, and the integrity of the call mix itself.

This divergence is not accidental. It is the predictable consequence of optimizing proxy metrics under pressure. When the proxy becomes the target, the system finds ways to hit the target without producing the outcome the target was supposed to represent.

4.2 The Strain Cliff

The jump from 2.4% to 99.6% of representatives in high/peak strain is not a gradual degradation—it is a phase transition. The pressure regime pushes the entire workforce across a threshold simultaneously. This suggests that contact center strain is not normally distributed across individuals; rather, the system has a critical point beyond which collective breakdown occurs.

This has direct implications for workforce management: by the time strain manifests in observable KPIs, the damage is already systemic. Leading indicators—particularly the strain distribution shape—are needed to detect the approaching cliff before the system crosses it.

4.3 Implications for AI-Driven Optimization

If an AI system were trained to optimize proxy FCR using this data, it would learn that the high-pressure condition is essentially equivalent to baseline—the target variable barely moves. It would not learn that the high-pressure condition produces worse true outcomes, more gaming, more fraud, higher burnout, and near-universal compliance risk. The training signal would be misleading, and the resulting model would perpetuate and potentially amplify the damage.

This validates the need for a System Integrity Index (SII) as a veto condition on AI optimization: before allowing any automated system to use proxy labels as training targets, the SII must confirm that those labels retain sufficient fidelity to reality.

5. Conclusion

The NovaWireless pressure experiment demonstrates that proxy KPI systems fail precisely when they are most trusted. Under high pressure, proxy FCR remains stable while every underlying reality metric deteriorates. The scenario mix shifts from clean interactions toward gaming and fraud. Representative strain undergoes a phase transition from manageable to universal. And none of this is visible on the standard KPI dashboard.

These findings argue for three operational changes: (1) governance frameworks that monitor the gap between proxy and true outcomes, not just the proxy itself; (2) strain distribution monitoring that can detect approaching phase transitions before they occur; and (3) mandatory integrity checks—such as the SII—as preconditions for any AI system that uses proxy metrics as optimization targets.

The green dashboard is not a sign of health. It is a symptom of a measurement system that has lost contact with reality.

Appendix: Dataset Specification

Dataset size: 60,000 synthetic call records (5,000 per month × 12 months). Generated using run_lab.py master orchestrator from the novawireless-lab repository. Master account ledger corrected to ~8% IMEI mismatch rate. Contract_proxy column renamed to billing_agreement_type. Representative pool: 250 synthetic agents with variable gaming propensity, burnout susceptibility, and skill profiles. Customer pool: 5,000 unique synthetic accounts with variable churn risk, patience, and trust baselines.